

AMUM — Agent Memory Update Module

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Independent Methodological Authority

OriginID: OOF-OID-MEM-AMUM-2026-06-08-0003

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Memory Governance Intelligence Architecture (MGIA™)

Operational Layer: Agent Memory Governance Layer

Governed Space: Agent Memory Update Integrity

Category: AI & Interpretation

Subcategory: Memory Update Governance

Type: Agent Memory Integrity Module

Parent Standard: Agent Memory Integrity Standard (AMIS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 8 June 2026

Compatibility: OOF Methodology OS · Agent Memory Integrity Standard (AMIS) ·

Memory Integrity Standard (MIS) · Memory Traceability Integrity Module (MTIM) ·

Memory State Integrity Module (MSIM) · Memory Evolution Integrity Standard (MEIS) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Agent Memory Update Module (AMUM) defines the structural conditions under which autonomous agents create, modify, revise, replace, enrich, correct, or remove memory objects while preserving traceability, accountability, consistency, reality alignment, governance oversight, and operational validity.

AMUM governs agent memory updates.

The module ensures that memory evolution remains controlled and explainable rather than becoming an uncontrolled process that gradually corrupts memory integrity.

A system satisfies AMUM only if:

- memory updates remain traceable
- update causes remain identifiable
- update history remains reconstructable
- update consequences remain assessable
- governance oversight remains preservable
- updates remain operationally valid

An agent that continuously modifies memory without accountability does not satisfy AMUM.

Module Operational Role

AMUM defines the memory-update governance layer of AMIS.

Its role is to preserve integrity throughout memory modification and evolution.

Module Operational Space

AMUM governs:

- memory updates
- memory modification
- memory correction
- memory enrichment
- memory replacement
- memory revision
- update accountability
- update governance

The module applies wherever agents alter memory states.

Module Function

The module applies wherever systems must preserve:

- accountable memory modification
- traceable memory evolution
- governance-valid updates
- reconstructable update history
- consistent memory development
- operationally reliable memory management

Its function is to ensure that memory changes remain visible, explainable, and governable.

Minimum Implementation Framework

1. Define the Memory Update Object

The organization must define which memory updates require governance.

This may include:

- context updates
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- knowledge updates
- preference updates
- operational updates
- environmental updates
- learning-based updates
- Human-AI memory updates
- autonomous memory modifications

2. Define Memory Update Conditions

The system must define the conditions under which updates remain valid.

This includes:

- traceability requirements
- accountability requirements
- consistency requirements
- validation requirements
- reality-alignment requirements
- governance-valid update conditions

3. Define Update Degradation Detection Logic

The system must define how update degradation is identified.

This may include:

- uncontrolled updates
- contradictory updates
- update-history loss
- memory corruption
- governance-blind modification
- reality-detached memory changes

4. Define Operational Response or Governance Logic

The system must define governance logic for update-integrity failures.

Governance response may include:

- update review
- rollback procedures
- validation activities
- governance intervention
- escalation
- update restrictions
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- update events
- modification history
- validation activities
- governance reviews
- intervention actions
- resulting memory states

An agent memory environment must not remain update-valid if materially significant memory modifications cannot be reconstructed, reviewed, or governed.

Use Case 1 — Learning Autonomous Agent

Scenario

An autonomous agent continuously updates memory based on operational experience, observations, and newly acquired knowledge.

Application

AMUM governs update traceability, modification accountability, and preservation of memory integrity throughout learning processes.

Result

The agent gains stronger learning reliability, improved governance visibility, and reduced exposure to memory corruption.

Use Case 2 — Enterprise AI Copilot

Scenario

An enterprise AI copilot continuously updates customer context, operational knowledge, organizational information, and interaction history.

Application

AMUM governs update accountability, memory consistency, and traceable memory evolution.

Result

The environment gains stronger memory reliability, improved transparency, and reduced exposure to uncontrolled memory modification.

Canonical Closing Statement

Agent Memory Update Module (AMUM) defines the structural conditions under which autonomous agents create, modify, revise, replace, enrich, correct, or remove memory objects while preserving traceability, accountability, consistency, reality alignment, governance oversight, and operational validity.

**Agent memory cannot remain trustworthy if updates become uncontrolled.
Memory update integrity therefore becomes a foundational
integrity condition of trustworthy autonomous memory systems.**

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